CEC 315 - Signals and Systems

Exam: Chapters 1 and 2 (Modified Version)

Spring 2026

Total: 100 points (4 problems × 20 pts + 5 multiple choice = 20 pts)

One problem per page. Show all your work.

Problem 1 (20 points)

System Properties

For each of the following systems, determine whether they are:

- **(i)** Memoryless or with memory
- **(ii)** Causal or non-causal
- **(iii)** Time-invariant or time-varying
- **(iv)** Linear or nonlinear

Justify each answer briefly.

(a) $y(t) = x(t-2)$

(b) $y(t) = t\,x(t)$

(c) $y[n] = x[n]\,x[n-1]$

Problem 2 (20 points)

Discrete-Time Convolution

Compute the convolution $y[n] = x[n] * h[n]$ for the following signals:

$$x[n] = \{1, 2, 2, 1\} \quad \text{for } n = 0, 1, 2, 3 \quad \text{(zero elsewhere)}$$
$$h[n] = \{1, 1, 1\} \quad \text{for } n = 0, 1, 2 \quad \text{(zero elsewhere)}$$

(a) State the range of n for which $y[n]$ is nonzero (output length).

(b) Compute $y[n]$ using the convolution sum $y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$.

Problem 3 (20 points)

Continuous-Time LTI Systems

(a) **Sifting property.** Evaluate the following integrals:

(i) $\displaystyle\int_{-\infty}^{\infty} (t^2 + 4t)\delta(t-2)\,dt$

(ii) $\displaystyle\int_{0}^{5} e^{-t}\delta(t-3)\,dt$

(b) **System properties.** For the impulse response $h(t) = e^{-4t}u(t)$:

(i) Is the system memoryless or with memory? Explain.

(ii) Is the system causal? Explain.

(iii) Is the system BIBO stable? *Show your work* by computing $\displaystyle\int_{-\infty}^{\infty} |h(t)|\,dt$.

(c) **Convolution property.** Using the identity $x(t) * \delta(t-t_0) = x(t-t_0)$, find and simplify: $\left[e^{-t}u(t)\right] * \delta(t-1)$.

Problem 4 (20 points)

Differential and Difference Equations

(a) **Continuous-time.** A causal LTI system is described by

$$\frac{dy(t)}{dt} + 5y(t) = 10\,x(t).$$

Find the impulse response $h(t)$.

(b) **Discrete-time.** A causal LTI system is described by the difference equation

$$y[n] = \frac{1}{2}y[n-1] + 3x[n].$$

(i) Use **iteration** to find $h[0]$, $h[1]$, and $h[2]$.

(ii) Write the general expression for $h[n]$.

(iii) Is this system stable? Justify briefly.

Part II: Multiple Choice (20 points total)

Instructions: Choose the **one** best answer for each question. No partial credit.
Points: 4 pts each.

1. An LTI system is completely characterized by:

(A) its step response

(B) its impulse response

(C) its frequency response

(D) its differential equation only

2. The convolution sum $y[n] = x[n] * h[n]$ is defined as:

(A) $\sum_{k=-\infty}^{\infty} x[k]h[n+k]$

(B) $\sum_{k=-\infty}^{\infty} x[k]h[n-k]$

(C) $\sum_{k=-\infty}^{\infty} x[n-k]h[k]$

(D) both (B) and (C) are correct

3. A continuous-time LTI system is BIBO stable if and only if:

(A) $h(t) = 0$ for $t < 0$

(B) $\int_{-\infty}^{\infty} h(t)\,dt < \infty$

(C) $\int_{-\infty}^{\infty} |h(t)|\,dt < \infty$

(D) $h(t)$ is bounded for all t

4. For the first-order differential equation $\frac{dy}{dt} + ay = bx$ with $a > 0$, the impulse response is:

(A) $h(t) = b\,e^{at}u(t)$

(B) $h(t) = b\,e^{-at}u(t)$

(C) $h(t) = \frac{b}{a}(1 - e^{-at})u(t)$

(D) $h(t) = \delta(t)$

5. A system with impulse response $h[n] = \delta[n - n_0]$ (shifted unit impulse) acts as:

(A) an accumulator

(B) an ideal delay of n_0 samples

(C) an amplifier

(D) an integrator

End of Exam